# Title: Vessel in the Void

# Domains: Digital Forensics + Reverse Engineering + Cryptography / Cryptanalysis

# Difficulty: 7/10

# Narrative

You are given a single file, `traffic.pcapng`, and nothing else. You open it and the noise stares back: handshakes, resets, idle HTTP, all perfectly ordinary—which is precisely what's wrong. Somewhere in this capture, one conversation refused to speak in bytes and chose to talk in pauses instead. One host pushes almost nothing, just a slow drip of packets like a metronome trying too hard not to be noticed. Follow it long enough and you'll find an attachment that shouldn't matter: a tiny gray square, blurred and boring, the kind of artifact you discard while chasing "real" exfiltration. You can stare at its pixels, brute-force its headers, or drown it in every stego tool on your machine; it will gladly waste your evening. The trick is that the frame and the rhythm are co-conspirators: the image only makes sense if you listen to the silence that delivered it. This challenge rewards the solver who questions layers, not the one who worships them.

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# Hints

**Hint 1**
The packets are talking, but the conversations between them gossip louder. Sometimes the gap says more than the payload.

**Hint 2**
There is only one flow patient enough to carry a secret. Ask which stream would still look suspicious if every application byte were stripped away.

**Hint 3**
If you uncover a tiny grayscale picture, treat it as a canvas, not the painting. Differences between neighboring pixels carry more truth than their absolute values.

**Hint 4**
Assume a single bit per packet in one TCP stream. The pattern does not stop after 256 packets; repetition is a feature, not a mistake.

**Hint 5**
Recover a repeating 256-bit sequence from inter-packet delays in the covert flow. Use that sequence as a symmetric key to unlock data hidden inside a small 48×48 PNG extracted from the same stream.